SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Metering Department	METERING & ENERGY ENGINEERING	**17th August 2026 – 21st August 2026**	**Week 11**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
Mechanical Meter Coupling, Component Seating & Real-Time Clock Battery Integration

WORK DONE DURING THE WEEK:
During this week in the Metering Assembly Station, I carried out mechanical coupling of 240 single-phase smart prepaid electricity meters. The core work done encompassed matching physical outer casing serials with electronic PCB barcodes, installing long-life 3.6V Li-SOCl2 RTC lithium backup batteries to safeguard non-volatile memory during power outages, seating 4G/GPRS IoT cellular communication daughterboards onto the motherboards, inspecting all internal terminal shunts, tightening casing screws using electric torque screwdrivers calibrated to 1.2 Nm to guarantee IP54 ingress protection against dust and moisture, and documenting the batch in the physical Coupling Records Ledger.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 17/08/2026	PCB & Enclosure Matching	Conducted 100% serial integrity audit matching front cover nameplate serials with electronic PCB barcodes. Inspected internal current transformer terminations.
Tuesday 18/08/2026	Internal Battery & LCD Integration	Installed long-life 3.6V Li-SOCl2 lithium backup batteries into PCB retention slots for real-time clock (RTC) retention. Mounted front intermediate housing ensuring clear LCD aperture seating.
Wednesday 19/08/2026	GPRS / Cellular Modem Coupling	Mounted 4G/GPRS communication daughterboards onto meter motherboards. Fastened internal whip antenna leads and validated SIM card tray retention mechanism.
Thursday 20/08/2026	Torque-Calibrated Screw Fastening	Secured outer transparent casing screws using calibrated electric torque screwdrivers (1.2 Nm) to ensure IP54/IP65 ingress protection against moisture and dust.
Friday 21/08/2026	Coupling Ledger Batch Reconciliation	Recorded 240 assembled meter units into the physical Coupling Records Ledger; audited serial batch ranges and routed units to the Post-Test station.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (METER COUPLING & COMPONENT LAYOUT HIERARCHY)

```
[Outer Transparent Cover] --> [Front Nameplate (Serial # Audit)]
   |--> [Sealing Gasket (IP54 Enclosure Ingress Protection)]
   |--> [Metering PCB Motherboard + LCD Display Aperture]
   |--> [3.6V Li-SOCl2 RTC Backup Battery] + [GPRS/4G IoT Daughterboard]
   |--> [Bi-Stable Magnetic Disconnect Latch Relay (100A Rating)]
   |--> [Base Terminal Housing + Micro-switch Cover Tamper Sensor]
```

STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Learned the critical importance of serial number reconciliation between physical casing and PCB firmware identity to avoid utility billing discrepancies in the Meter Management System (MMS)."

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COMPUTER WAREHOUSE GROUP (CWG) PLC
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

TRAINING RECORD
CWG Infrastructure & Engineering

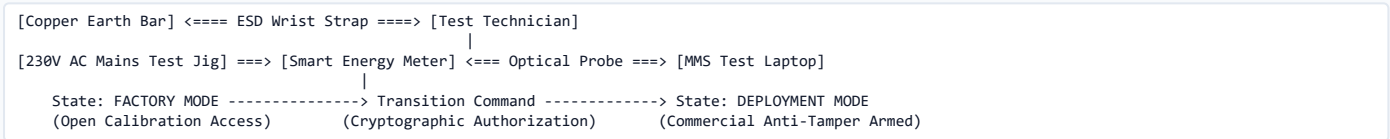
SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Metering Department	METERING & ENERGY ENGINEERING	**24th August 2026 – 28th August 2026**	**Week 12**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
ESD Grounding Setup, Mains Energization, Keypad Verification & Factory-to-Deployment Mode Switching

WORK DONE DURING THE WEEK:
During this week in the Post-Test Inspection Station, I carried out functional testing and operational mode parameterization of coupled smart meters. Work done included setting up anti-static Electrostatic Discharge (ESD) wrist straps grounded to the common copper earth bar (< 1 MOhm) to prevent gate oxide puncture on CMOS/MCU chips, energizing meters at 230V AC / 50Hz nominal mains, sequentially testing all tactile keys on the 12-key elastomer keypad with diagnostic shortcodes (#001#, #002#, #003#), interfacing infrared optical read-heads to execute proprietary Meter Management Software (MMS) scripts that transitioned meters from Factory Mode to Deployment Mode, and generating 20-digit STS tokens to test relay disconnector tripping.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 24/08/2026	ESD Workstation Grounding Setup	Inspected ESD wrist straps, verified < 1.0 MOhm resistance path to common copper earth bar, and initiated static-safe post-test workbench procedures.
Tuesday 25/08/2026	Mains Energization & LCD Boot Diagnostics	Connected coupled smart meters to 230V AC / 50Hz bench power supply; checked power-on self-test (POST) sequences, LCD segment illumination, and relay click.
Wednesday 26/08/2026	Elastomer Keypad Verification	Sequentially entered diagnostic test shortcodes (#001# for Total Energy, #002# for Instantaneous Voltage, #003# for Current) to confirm audio-tactile elastomer keypad functionality.
Thursday 27/08/2026	MMS Optical Firmware Parameterization	Coupled infrared optical probes to meter front read-heads; used Meter Management Software (MMS) to switch operational mode from Factory Mode to locked Deployment Mode.
Friday 28/08/2026	Relay Disconnector Operation & Token Acceptance	Generated 20-digit STS test recharge credit tokens via test generator; loaded tokens via keypad to confirm token acceptance and internal latch relay disconnector trip/close.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (POST-TEST BENCH SETUP & MMS MODE STATE MACHINE)



STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Mastered the use of proprietary Meter Management Software (MMS) and optical read-heads. Understood how ESD grounding prevents latent dielectric breakdown in microcontroller gates."

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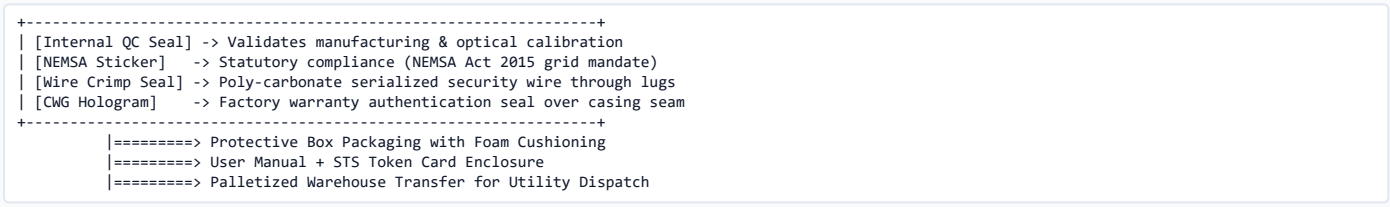
SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Metering to IT Systems Handover	METERING & ENERGY ENGINEERING	**31st August 2026 – 4th September 2026**	**Week 13**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
NEMSA Regulatory Compliance Audit, Tamper-Evident Wire Sealing & Palletized Warehouse Transfer

WORK DONE DURING THE WEEK:
During this week at the Final Packaging & Regulatory Certification Station, I performed statutory quality compliance checks and anti-tamper sealing on 300 tested meters. Work done included conducting visual surface audits and cleaning grease/fingerprint residue with anti-static solution, affixing internal QC Pass stickers over enclosure joints, applying statutory Nigerian Electricity Management Services Agency (NEMSA) regulatory certification labels pursuant to the NEMSA Act 2015, threading stainless steel sealing wire through enclosure sealing lugs, crimping serialized poly-carbonate NEMSA seals, affixing CWG holographic warranty security seals, boxing units with user manuals, and reconciling packaging ledgers.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 31/08/2026	Finished Cosmetic & Housing Audit	Inspected 300 tested meters for cosmetic flaws, stripped terminal screws, or missing terminal cover screws. Cleaned surface grease using anti-static fluid.
Tuesday 01/09/2026	QC Pass & Statutory NEMSA Sticker Application	Affixed internal QC Pass guarantee stickers over enclosure seam joints; applied official statutory NEMSA (Nigerian Electricity Management Services Agency Act 2015) compliance labels.
Wednesday 02/09/2026	Serialized NEMSA Poly-carbonate Wire Sealing	Threaded high-tensile stainless steel sealing wire through meter terminal lugs and casing lugs; crimped serialized NEMSA tamper-evident seals using calibrated sealing pliers.
Thursday 03/09/2026	CWG Holographic Sealing & User Manual Insertion	Affixed CWG Plc corporate holographic tamper-evident warranty seals; inserted user operating manuals, STS quick-start guides, and wiring schematics into product cartons.
Friday 04/09/2026	Packaging Ledger Audit & Departmental Handover	Completed batch ledger documentation reconciling 300 meter serials with NEMSA seal serials; transferred palletized batch to warehouse store. Prepared for rotation to IT Support.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (NEMSA & CWG TAMPER-EVIDENT MULTI-TIER SEALING SCHEME)



STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Learned the legal and technical necessity of statutory NEMSA verification for grid-connected metering equipment in Nigeria, and how multi-tier physical security seals deter meter tampering."

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STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

TRAINING RECORD
CWG Infrastructure & Engineering

SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Enterprise IT Infrastructure	IT SUPPORT & PROXIMITY	**14th September 2026 – 18th September 2026**	**Week 14 (IT Week 1)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
IT Proximity Helpdesk Orientation, Ticketing SLA Review & POS Retail Suite Architecture Kickoff

WORK DONE DURING THE WEEK:
During this week in the IT Support & Proximity Unit, I commenced enterprise helpdesk operations and software project planning. Work done included participating in orientation on the IT Service Management (ITSM) ticketing tool and SLA escalation matrices, resolving 8 corporate desktop support tickets (printer spooler crashes, Microsoft 365 Outlook profile synching, and local IP mapping), participating in the software architecture kickoff meeting for the CWG Retail Suite (Point of Sale & Inventory), drafting the relational Entity-Relationship Diagram (ERD), scaffolding the Laravel 12 backend with Sanctum authentication, and setting up the Vite-powered React frontend repository on GitHub.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 14/09/2026	Proximity Unit Orientation & ITSM Workflow	Reported to IT Support & Proximity Unit. Received orientation on enterprise IT service desk operations, SLA priority matrices (Severity 1 to 4), and incident ticketing platforms.
Tuesday 15/09/2026	Corporate User Incident Handling	Resolved 8 Level-1 user tickets: diagnosed corporate printer spooler crashes, configured local network shared printers, and assisted staff with Microsoft 365 Outlook profile sync.
Wednesday 16/09/2026	CWG Retail Suite Architecture & ERD Review	Participated in software development kickoff meeting for CWG Retail Suite (POS & Inventory). Reviewed system requirements and drafted relational Entity-Relationship Diagram (ERD).
Thursday 17/09/2026	Laravel 12 API Scaffold & Sanctum Setup	Installed Laravel 12 backend framework; configured Laravel Sanctum tokenized authentication and structured database connection parameters for MySQL in .env.
Friday 18/09/2026	Initial React Frontend Scaffolding & Git Tracking	Scaffolded modern React frontend client using Vite; initialized Git version control repository on GitHub with branch protection rules.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (ITSM ESCALATION & CLIENT-SERVER SYSTEM ARCHITECTURE)

```
[User Incident / Ticket] ---> [Level-1 Proximity Dispatch] ---> [Level-2 Desktop Support]
                                                                        |
                                  [CWG Retail Suite Web Client (React 19)]
                                                  | (RESTful JSON via HTTP)
                                  [Laravel 12 API Server + Sanctum Token Auth]
                                                  | (Eloquent ORM)
                                  [MySQL Relational Database Engine (Port 3306)]
```

STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Gained clear perspective on enterprise helpdesk operations and corporate incident escalation. Successfully initialized the foundational full-stack decoupled architecture for CWG Retail Suite."

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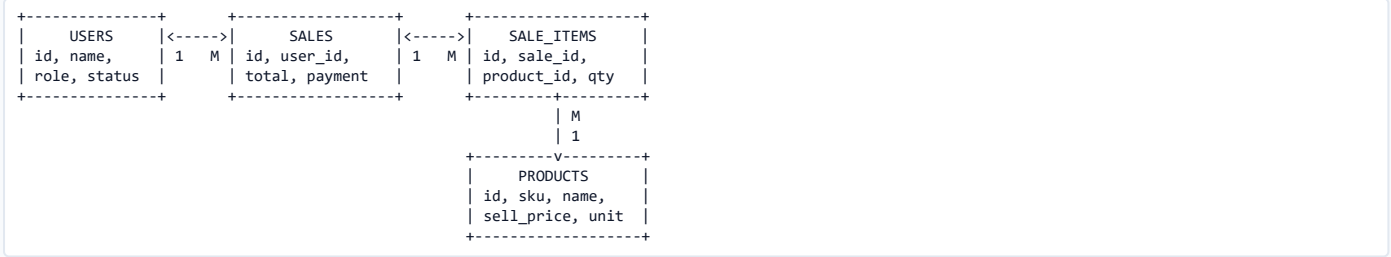
SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Infrastructure & Systems Development	SOFTWARE & IT SYSTEMS	**21st September 2026 – 25th September 2026**	**Week 15 (IT Week 2)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
Enterprise VPN/MFA Support, Database Schema Normalization & Secure RESTful Authentication APIs

WORK DONE DURING THE WEEK:
During this week across IT Proximity and Software Development, I resolved enterprise user identity issues and developed core backend APIs. Work done included configuring secure remote VPN tunnels (Zscaler Private Access and Cisco AnyConnect) for remote executives, enrolling multi-factor authentication (MFA) devices via Microsoft Authenticator, designing and running normalized database migrations in Third Normal Form (3NF) for users, products, sales, and line items, writing tokenized authentication controllers using Bcrypt password hashing, and disassembling two corporate laptops to upgrade failing mechanical HDDs to high-speed 512GB M.2 NVMe SSDs.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 21/09/2026	Remote VPN & MFA Configuration	Assisted corporate staff with Zscaler / Cisco AnyConnect secure remote VPN configurations and registered Microsoft Authenticator Multi-Factor Authentication (MFA) tokens.
Tuesday 22/09/2026	Database Migrations & Relational Schema (3NF)	Designed and executed Laravel migrations in 3rd Normal Form: `users`, `products`, `sales`, `sale_items`, and `stock_adjustments` tables with foreign key constraints.
Wednesday 23/09/2026	Sanctum Token Authentication & Breeze Integration	Implemented user login, password hash verification (Bcrypt), and token issuance endpoints (`POST /api/login`) with role validation attributes.
Thursday 24/09/2026	Hardware RAM/SSD Laptop Diagnostic	Diagnosed 2 malfunctioning corporate HP laptops: replaced faulty DDR4 SODIMM RAM modules, upgraded failing HDD to 512GB NVMe SSD, and verified BIOS hardware recognition.
Friday 25/09/2026	Product Creation API & Route Testing	Developed `ProductController.php` with validation rules; created API endpoint for product catalogue management and tested JSON responses via automated test suite.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (DATABASE RELATIONAL SCHEMA (3NF) TOPOLOGY)



STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Understood the practical intersection between hardware maintenance (SSD/RAM staging) and software data integrity (database normalization and token security)."

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SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Enterprise Networking & Workstation Maintenance	IT SUPPORT & NETWORKING	**28th September 2026 – 2nd October 2026**	**Week 16 (IT Week 3)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
Structured Cabling, Patch Panel Termination & Network Continuity Diagnostics

WORK DONE DURING THE WEEK:
During this week in the Network Infrastructure & Data Center Unit, I carried out structured cabling and local area network diagnostics. Work done included performing an audit of server rack patch panels on the 2nd floor, tracing unmanaged drop cables, terminating 12 new solid-copper Cat6 cables into 24-port patch panels following the TIA/EIA-568B pinout standard, conducting cable continuity and near-end crosstalk tests with an electronic wire tester, assisting Senior Engineers with switch port patching across departmental VLANs (VLAN 10 Admin, VLAN 20 Engineering), and deploying Aruba enterprise Wi-Fi access points to improve coverage in executive meeting spaces.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 28/09/2026	Data Center Patch Panel Audit	Audited server rack patch panels on the 2nd floor data room; re-labeled Cat6a drop cables and documented port assignments to Cisco Catalyst 2960 switches.
Tuesday 29/09/2026	Cat6 Structured Cabling & RJ-45 Crimping	Terminated 12 new Cat6 solid copper patch cables according to the TIA/EIA-568B standard; verified pinout continuity, wire-map, and near-end crosstalk with digital cable tester.
Wednesday 30/09/2026	Cisco Switch VLAN Patching & Trunking	Assisted Senior Network Engineer in patching workstation ports to designated Departmental VLANs (VLAN 10: Admin, VLAN 20: Engineering, VLAN 30: Guest Wi-Fi).
Thursday 01/10/2026	Public Holiday: National Independence Day	Statutory National Holiday (Nigeria Independence Day). Conducted scheduled remote server health checks and UPS power status monitoring.
Friday 02/10/2026	Wireless Access Point (AP) Deployment	Mounted and configured Aruba Enterprise Wi-Fi Access Points; measured RF signal coverage and resolved channel interference in the executive conference boardroom.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (TIA/EIA-568B COLOR CODE & ENTERPRISE SWITCH PATCHING)

```
TIA/EIA-568B Pinout Standard:
Pin 1: White/Orange | Pin 2: Orange      | Pin 3: White/Green | Pin 4: Blue
Pin 5: White/Blue   | Pin 6: Green       | Pin 7: White/Brown | Pin 8: Brown
                      |
[Wall Data Outlet RJ-45] ===Cat6 Drop===> [24-Port Patch Panel] ===Patch Cord===> [Cisco Switch (VLAN Tagged)]
```

STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Mastered precision RJ-45 crimping under TIA/EIA-568B. Gained real-world appreciation for structured cable dressing and cable management in high-density rack enclosures."

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SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Identity & Access Management	ACTIVE DIRECTORY ADMINISTRATION	**5th October 2026 – 9th October 2026**	**Week 17 (IT Week 4)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
Active Directory Domain Services, User Provisioning, OUs & Group Policy Object (GPO) Deployment

WORK DONE DURING THE WEEK:
During this week in Systems Administration, I performed user identity management and policy enforcement on Windows Server 2022 Active Directory Domain Services (AD DS). Work done included provisioning new employee accounts in Active Directory Users and Computers (ADUC), categorizing user objects into departmental Organizational Units (OUs), creating and linking Group Policy Objects (GPOs) for screen timeouts, minimum password complexity (12 characters), and USB mass storage write protection, joining 6 corporate Dell laptops to the `cwg.local` domain, verifying DNS and Kerberos authentication tickets, and executing PowerShell scripts to audit inactive user accounts.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 05/10/2026	Active Directory User Provisioning	Created new corporate user accounts in Windows Server 2022 Active Directory Users and Computers (ADUC); assigned complex password policies and primary email aliases.
Tuesday 06/10/2026	Organizational Units (OUs) & Departmental Groups	Structured directory tree hierarchy: segregated user objects into nested OUs (`CWG-Staff`, `Metering-Dept`, `IT-Operations`, `Finance`); assigned security group memberships.
Wednesday 07/10/2026	Group Policy Objects (GPOs) Configuration	Configured and linked Domain GPOs: enforced 15-minute screen lock with password, restricted USB mass storage write access, and mapped departmental network drives (`Z:\Enterprise_Data`).
Thursday 08/10/2026	Domain Joining & Kerberos Trust Diagnostics	Joined 6 new corporate Dell Latitude laptops to domain `cwg.local`; verified DNS server lookups, computer account registration, and Kerberos ticket generation.
Friday 09/10/2026	Audit of Inactive AD Accounts & Password Resets	Executed PowerShell AD audit scripts to identify user accounts inactive for > 60 days; processed 14 self-service password reset requests from enterprise staff.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (ACTIVE DIRECTORY HIERARCHY & GPO INHERITANCE FLOW)

```
Domain: cwg.local
  |--> OU: Corporate Head Office
         |--> OU: IT-Operations -------> [GPO: Admin Access & Remote Management]
         |--> OU: Metering-Division ---> [GPO: Optical Testing Hardware Drivers]
         |--> OU: Commercial Staff ----> [GPO: USB Write Block & Inactivity Lock]
                 |--> Users: [staff1, staff2] (Members of 'Domain Users' + 'POS-Operators')
```

STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Gained in-depth hands-on experience managing Windows Server Active Directory. Understood how GPOs standardize security policies across hundreds of networked endpoints simultaneously."

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STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) — TECHNICAL LOGBOOK

TRAINING RECORD
CWG Infrastructure & Engineering

SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Core POS Engine & Inventory Architecture	SOFTWARE ENGINEERING	**12th October 2026 – 16th October 2026**	**Week 18 (IT Week 5)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
Product Inventory Normalization, Stock Adjustments Engine & Role-Based Middleware Security

WORK DONE DURING THE WEEK:
During this week in the Software Engineering team, I built the inventory tracking and access control layers of the CWG Retail Suite. Work done included refactoring ProductController to support Route Model Binding and soft deactivation via `is_active` flags, decoupling stock quantities into an independent `inventories` table, developing `StockAdjustmentController.php` with audit reason tagging (`restock`, `damaged`, `audit`), implementing atomic database transactions to guarantee stock consistency during concurrent operations, authoring PHPUnit automated feature tests, and writing role-based middleware that strictly bars cashiers from viewing cost prices, supplier margins, and administrative settings.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 12/10/2026	Product Deactivation & Route Model Binding	Refactored `ProductController` to use Laravel Route Model Binding; added soft-delete `is_active` flag to archive products safely without breaking historical sales data.
Tuesday 13/10/2026	Decoupled Inventories Table Migration	Separated stock quantity into dedicated `inventories` table with 1-to-1 product relationship; migrated existing product inventory data seamlessly.
Wednesday 14/10/2026	Stock Adjustment Controller & Audit Routing	Implemented `StockAdjustmentController.php` supporting stock audit reasons (`damaged`, `restock`, `inventory_audit`); built transaction logging for stock movements.
Thursday 15/10/2026	Multi-Product Automated Feature Tests	Authored Pest/PHPUnit automated test suites (`StockAdjustmentTest.php`) verifying inventory increments, boundary conditions, and negative stock prevention.
Friday 16/10/2026	Role Middleware & Cashier Restrictions	Created custom `EnsureUserHasRole` middleware; ensured cashiers are strictly barred from viewing product cost prices, margin data, or store settings.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (INVENTORY AUDIT LOGGING & ROLE GATE STATE MODEL)

```
Request -> [Role Middleware Check]
            |--> Role == Cashier  ---> Returns: sell_price, sku, name (cost_price HIDDEN)
            |--> Role == Owner/Mgr --> Returns: cost_price, profit_margin, full_audit
                                  |
[Stock Adjustment] ---> [DB Transaction Lock] ---> [Adjust Inventory Qty] ---> [Log Audit Event]
```

STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Learned professional test-driven development (TDD) using automated PHPUnit assertions. Successfully protected sensitive business financial metrics using server-side role middleware."

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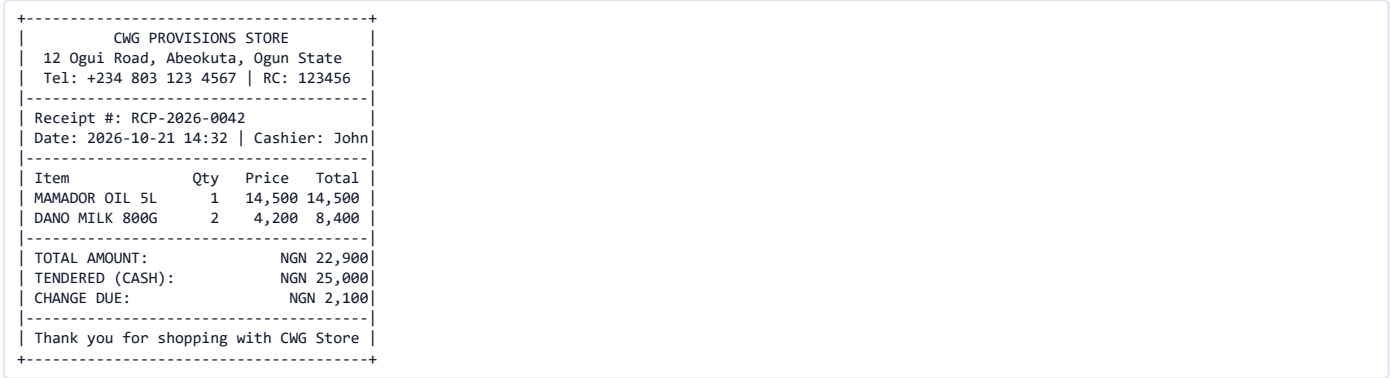
SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
POS Checkout, Thermal Printing & Sales Engine	SOFTWARE ENGINEERING	**19th October 2026 – 23rd October 2026**	**Week 19 (IT Week 6)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
POS Checkout Interface, Barcode SKU Search, 80mm Thermal Receipt Generator & Sales History

WORK DONE DURING THE WEEK:
During this week in Software Development, I focused on client-side checkout and peripheral hardware integration. Work done included engineering the Point of Sale (POS) user interface in React with real-time barcode scanning, search auto-complete, and cart quantity capping, building an atomic cash tender dialog with automated change calculation, styling an 80mm browser-native thermal receipt print layout using CSS `@media print` rules for thermal printers, creating a paginated Sales History table with multi-parameter date range filters, seeding 14 authentic Nigerian retail products, and building an inventory Restock intake station.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 19/10/2026	POS Interface Grid & Real-Time Cart State	Built React POS component with live category filtering, rapid barcode/SKU search, responsive product grid, and stock-capped cart line item management.
Tuesday 20/10/2026	Atomic Cash Checkout & Change Calculator	Engineered checkout settlement modal supporting Cash, Card, and Bank Transfer tender options with automated change calculation and transactional concurrency locking.
Wednesday 21/10/2026	Browser-Native 80mm Thermal Receipt Engine	Implemented browser-native 80mm thermal receipt print stylesheet (`@media print`); styled store header, cashier credentials, line items, VAT, and barcode footer.
Thursday 22/10/2026	Sales History Table & Date Range Filter Presets	Created sales audit table with filtering by payment method and date presets (Today, 7D, 30D); integrated receipt reprint modal.
Friday 23/10/2026	Database Seeding & Restock Intake Station	Seeded 14 authentic Nigerian retail products with images; developed Restock & Adjustment station with reason-tagged adjustments and visual stock badges.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (80MM POS THERMAL RECEIPT LAYOUT STRUCTURE)



STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Built an authentic, high-speed retail checkout flow with strict inventory synchronization. Designed pixel-perfect 80mm thermal receipt layouts that print without browser margins."

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SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Financial Reports & High-Performance Optimization	SOFTWARE ENGINEERING	**26th October 2026 – 30th October 2026**	**Week 20 (IT Week 7)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
Financial Reports Engine, SQL B-Tree Indexing, React-Toastify Popups & Activity Audits

WORK DONE DURING THE WEEK:
During this week in Software Engineering, I executed critical database query optimizations and built enterprise analytical reporting tools. Work done included eliminating sequential N+1 query loops across sales lookups, applying compound B-Tree indexes on `sales(created_at, payment_method)` and projecting specific SQL columns to dramatically accelerate database response, developing the 5-tab Reports module (Financial Sales, Inventory Valuation, Stock Health, Product Velocity, and User Staff Audits) with CSV and PDF export, organizing 30-day adaptive sales charts, integrating React-Toastify notifications, and incorporating real-world metering work instructions into the IT technical report.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 26/10/2026	Database Performance & SQL B-Tree Indexing	Eliminated sequential N+1 query loops; added compound B-Tree indexes on `sales(created_at, payment_method)` and projected columns at SQL level for instant response times.
Tuesday 27/10/2026	Financial Analysis Layout & 5 Report Subtabs	Constructed comprehensive Reports module: Financial Sales, Inventory Valuation, Stock Level Deficits, Product Velocity, and User Staff Audits with CSV/PDF export.
Wednesday 28/10/2026	Adaptive 30-Day Sales Chart & SweetAlert2 Modals	Organized 30-day/90-day daily sales revenue chart with collision-free interval ticks; replaced old browser alerts with modern themed dialogs and React-Toastify notifications.
Thursday 29/10/2026	Staff Activity Audit Log & Admin Layer Isolation	Organized admin-only pages under dedicated `pages/admin` directory; added staff event timeline capturing sales transactions, logins, and inventory restock operations.
Friday 30/10/2026	IT Technical Report Metering Instructions Update	Incorporated physical manufacturing work instructions from the metering factory (Coupling, Post-Test ESD Safety, and NEMSA Packaging) into the comprehensive IT Report.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (SQL QUERY OPTIMIZATION: BEFORE VS AFTER B-TREE INDEXING)

```
BEFORE OPTIMIZATION:
  Query: SELECT * FROM sales WHERE created_at BETWEEN ? AND ?
  Execution: Full Table Scan (ALL) -> 5,000+ rows traversed -> High Server Latency (1,800ms)

AFTER B-TREE INDEXING:
  Schema: INDEX sales_created_payment_idx (created_at, payment_method)
  Execution: Range Index Scan (range) -> Directly fetches matched pointer rows -> Sub-10ms response
```

STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Learned the deep technical mechanics of database query optimization, compound B-tree index execution plans, and architectural modularity in enterprise web development."

INDUSTRY SUPERVISOR
Signature & Date Stamp

INSTITUTIONAL SUPERVISOR
SIWES Coordinator Signature

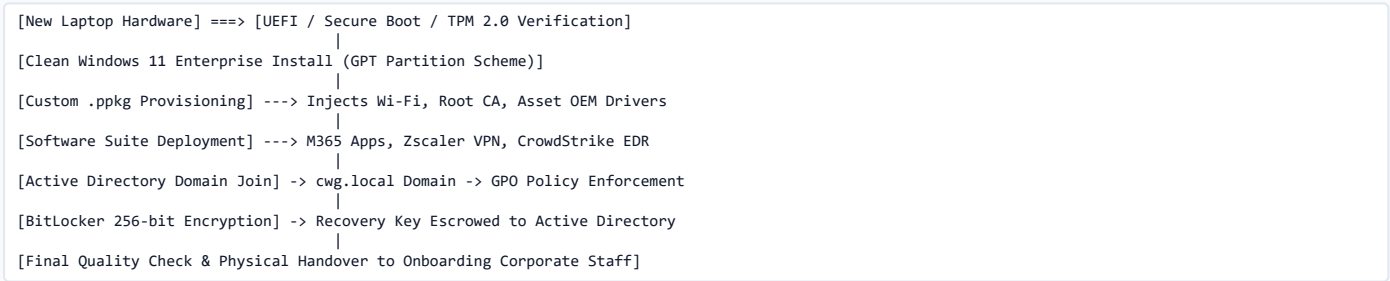
SECTION / DEPARTMENT	OPERATIONAL UNIT	PERIOD OF TRAINING	LOGBOOK WEEK
Enterprise IT Operations	IT SUPPORT & SYSTEMS DEPLOYMENT	**2nd November 2026 – 6th November 2026**	**Week 21 (IT Week 8)**

PROJECT / MAIN JOB ASSIGNED DURING THE WEEK:
Laptop Hardware Staging, Windows OS Provisioning (.ppkg), Essential Software Deployment & Active Directory Domain Integration

WORK DONE DURING THE WEEK:
During this week in the IT Support & Systems Staging Lab, I conducted complete lifecycle preparation and deployment of enterprise laptop hardware for onboarding staff. Work done included unboxing and inspecting 5 Dell Latitude 5440 laptops, configuring UEFI BIOS settings (Secure Boot, TPM 2.0, battery management), performing clean Windows 11 Enterprise installation over GPT partition tables, executing automated staging via Windows Configuration Designer provisioning packages (.ppkg) that injected Wi-Fi profiles, root certificates, and OEM drivers, deploying the enterprise application stack (Microsoft 365, Zscaler VPN, CrowdStrike Falcon), joining devices to the `cwg.local` Active Directory domain, escrowing BitLocker keys to AD, and completing employee handover.

DAY & DATE	NATURE OF WORK DONE	DAILY DETAILED TECHNICAL ACTIVITIES & OPERATIONS PERFORMED
Monday 02/11/2026	New Laptop Unboxing & Hardware POST Verification	Unboxed batch of 5 brand new corporate enterprise laptops (Dell Latitude 5440); verified battery health, verified UEFI BIOS firmware settings, secure boot, and TPM 2.0 status.
Tuesday 03/11/2026	Windows 11 Enterprise Clean OS Installation	Configured UEFI bootable flash drives with official Windows 11 Enterprise media; partitioned NVMe SSD storage (GPT scheme) and completed OS base installation.
Wednesday 04/11/2026	Automated Staging via Provisioning Packages (.ppkg)	Connected corporate provisioning package (.ppkg) generated with Windows Configuration Designer: automatically injected Wi-Fi profiles, corporate root CA certificates, and OEM drivers.
Thursday 05/11/2026	Essential Enterprise Software & Security Suite Installation	Installed corporate software stack: Microsoft 365 Enterprise Apps (Word, Excel, Outlook, Teams), Zscaler Private Access VPN, CrowdStrike Falcon EDR antivirus, and Google Chrome Enterprise.
Friday 06/11/2026	Domain Joining, Active Directory Staging & User Profile Handover	Joined newly staged laptops to `cwg.local` Active Directory domain; moved computer objects into `OU=Workstations,OU=CWG-Staff`; tested domain login, BitLocker drive encryption escrow to AD, and handed over to onboarding employee.

TECHNICAL DIAGRAM / SCHEMATIC / ARCHITECTURAL WORKFLOW (COMPLETE LAPTOP STAGING & ACTIVE DIRECTORY ONBOARDING WORKFLOW)



STUDENT COMMENTS & KEY LEARNINGS FOR THE WEEK
"Acquired comprehensive practical mastery in end-to-end enterprise laptop staging, automated provisioning with .ppkg packages, essential enterprise security deployment, and Active Directory domain enrollment."

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